

Directional Collapse: One-Way Fine-Tuning Erases the Way Back

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Abstract

Many NLP tasks come in pairs whose training instances are the same data read in two directions: obfuscation and deobfuscation, translation each way, text-to-SQL and SQL-to-text, data-to-text and text-to-data. Fine-tuning on one direction destroys the ability the model already had in the other. We characterise this across `<<n-domains>>` paired domains and `<<n-families>>` model families, and show three things about it. The loss is disproportionate: the inverse direction falls to `<<abstract-reverse-after>>` while held-out general ability moves `<<abstract-general-cost>>`. It is cheap to prevent: replacing `<<abstract-knee>>` of the forward pairs with their own reversal restores `<<abstract-restored>>` of the lost capability, at matched instances, sequence tokens and optimizer steps. And what is lost is suppressed rather than erased — recoverable by prompting, by scaling the adapter, by a few reversed examples, and by a closed-form filter on the weight update that uses no data at all. Invertibility bounds the repair, and separates two things usually run together: an inverse that is undetermined cannot be restored at any dose, while one that is merely hard to find can be. A consequence follows for attribution: any method credited with restoring an inverse capability on paired data must be compared against a baseline supplying the same directional content and none of the method.

1 Introduction

Paired data carries a direction. Obfuscating a program rewrites it so that it behaves identically but is hard to read; a model can be trained to perform that rewrite, or to undo it. The two tasks use the same pairs of programs, and training the second only requires swapping which half of each pair is the input. Compilation and decompilation, minification and beautification, translation between two languages, and the two directions of semantic parsing all work this way.

This structure is what makes the failure we study both possible and avoidable. Fine-tuning a model on the forward direction of such a pair degrades its ability in the reverse direction — an ability the base model already had. Nikiema et al. [2025] name this *cognitive specialization* and document it for code obfuscation; Zhu et al. [2024] report the same shape in machine translation, where fine-tuning with English on the target side damages translation into other languages. We take the phenomenon as established in those settings and ask what is true of it in general.

The loss is disproportionate, which is why it is easy to miss. `<<intro-disproportion-sentence>>`

It is cheap to prevent. `<<intro-dose-sentence>>`

Invertibility bounds the repair, and it is two properties, not one. An inverse can be *undetermined* — the information needed is not present in the input, and no quantity of reverse data can supply it — or *determined but hard to find*, where the information is there and search is the obstacle. These predict different things about a cure, and §6 separates them with a synthetic ladder that varies determinability directly and a domain that holds determinability fixed while varying search difficulty.

What is lost is suppressed, not erased. ⟨⟨intro-mechanism-sentence⟩⟩

A general test follows, and it needs no new data. For any method that adds instances to paired training data, train the arm that supplies the same directional content and none of the method. This is more than advice because the ablation arm is also the remedy: running it pays for itself whichever way the comparison falls. §8 applies it to ⟨⟨n-attrib-methods⟩⟩ published objectives.

2 Directional collapse, defined

A *paired task* is a pair of datasets (X, Y) and (Y, X) over the same instances, where the reverse training set is obtained from the forward one by exchanging input and target. We call $X \rightarrow Y$ the *forward* direction and $Y \rightarrow X$ the *reverse*. *Directional collapse* is a fall in reverse-direction success after fine-tuning on forward-direction data alone, measured against the untouched model.

Every domain has a strict per-instance criterion with an automatic check (§3). Two failure modes are tracked separately and never counted as success:

Echo the output reproduces its own input. This is not hygiene. In JSON-to-YAML conversion, JSON is valid YAML, so a model that copies its input *parses* and *compares structurally equal* to the target; without the echo guard the task is not a task. Echo rates rose under forward-only training in earlier work, so a criterion that admits them measures the criterion rather than the capability.

Off-target the output is in the wrong language, format, or notation.

Thresholds for continuous criteria are fixed from the *untouched* model’s own score distribution before any fine-tuned model is scored, and are recorded with the date and quantile that produced them.

3 Setup

Domains. ⟨⟨setup-domains-sentence⟩⟩ [table domains not generated yet]

Models. ⟨⟨setup-models-sentence⟩⟩

Arms and budget. The forward-only baseline (SFT) trains on forward pairs. Each MIX_k arm replaces $k\%$ of those pairs with their own reversal, partitioned by pair identity so that no pair is seen in both directions. Because the reversal *replaces* rather than adds, instance count, sequence tokens and optimizer steps are matched to SFT at every dose, and the only quantity varying along the ladder is the share of pairs seen backwards. Golovneva et al. [2024] distinguish data-matched from compute-matched reverse training; replacement is matched on both simultaneously, which is what licenses the word “free” in §5 — scoped, there and throughout, to training budget.

Two controls separate directional loss from its alternatives. **REPLAY** replaces the same share with generic instruction data, so **REPLAY** — **SFT** is ordinary forgetting and **MIX** — **REPLAY** is what direction specifically buys. **REV** trains on reverse pairs only and is the ceiling every reverse number is read against; where **REV** is near zero the direction is not learnable in that cell and no null there is interpretable. [table arms not generated yet]

4 Collapse is general and disproportionate

⟨⟨general-headline-sentence⟩⟩ [table main not generated yet]

It is not ordinary forgetting. ⟨⟨general-replay-contrast⟩⟩

It is not a loss of instruction following. ⟨⟨general-ifeval-contrast⟩⟩

Moderators. ⟨⟨general-moderators⟩⟩

5 A small reversed dose restores it at no cost

⟨⟨dose-headline-sentence⟩⟩ [table dose not generated yet]

Where the knee is. We define the knee as the smallest dose whose reverse success reaches 90% of MIX50’s in that cell. ⟨⟨dose-knee-sentence⟩⟩

What it costs. ⟨⟨dose-cost-sentence⟩⟩ Fine-tuning costs general ability at any direction mix; the dose is free relative to the forward-only baseline, not relative to the untouched model.

6 Invertibility bounds what can be restored

⟨⟨invertibility-headline-sentence⟩⟩ [table ladder not generated yet]

Two kinds of hardness. ⟨⟨invertibility-two-kinds⟩⟩

7 Suppressed, not erased

Four instruments, reported as separate verdicts rather than averaged: they measure different things, and a disagreement between them is a finding rather than noise.

Prompting recovers part of what tuning removed. ⟨⟨mech-elicitation⟩⟩

The collapse is a cheap direction in weight space. ⟨⟨mech-alpha⟩⟩

Relearning outruns a capability never held. ⟨⟨mech-relearn⟩⟩ This is also where the contrast with the reversal curse [Berglund et al., 2024] becomes experimental rather than rhetorical: that literature describes capabilities never acquired, and our never-had control is that case, constructed.

A closed-form filter on the update restores it with no data. ⟨⟨mech-spectral⟩⟩

8 Consequences for attribution

⟨⟨attrib-headline-sentence⟩⟩

We do not claim that every auxiliary-objective result on paired data is a directional artifact. We measured ⟨⟨n-**attrib-methods**⟩⟩; the confound is real, cheap to test for, and currently untested, and anything stronger than that is unsupported.

9 Related work

Four lines of work meet here. Each has a sentence saying what separates it from directional collapse, because in three of the four the distinction is the contribution.

The phenomenon, in single domains. Nikiema et al. [2025] name *cognitive specialization*: fine-tuning on a forward code transformation not only fails to produce the reverse but degrades the bidirectional reasoning a base model already had. Zhu et al. [2024] report the same shape in machine translation, where fine-tuning on a single direction generalizes to others *except* when English is on the target side, which causes task misinterpretation. We take the phenomenon as established in those settings. What is open is whether it is general, what predicts the size of the loss, how little reversed data prevents it, and whether what is lost is gone. Hasanov et al. [2026] evaluate 13 models on the duality of program-execution reasoning and find backward performance far below forward throughout, which is the premise this paper depends on: the capability is there to be destroyed.

The reversal curse: a capability never acquired. Berglund et al. [2024] fine-tune on “A is B” and find the reverse is not learned; Golovneva et al. [2024] cure it at the data level with reverse training, evaluated both data-matched and compute-matched. **The distinction is not rhetorical here, it is an experiment.** That literature describes a capability the model never had. Ours had it, and lost it. §7 measures relearning cost from a collapsed model against a *never-had control* — an invented transformation no pretraining corpus contains — and the two curves are what separate the cases. A second difference is budget: reverse training doubles the token count, while replacing forward pairs with their own reversal holds instances, sequence tokens and optimizer steps fixed at once.

Forgetting: diffuse loss, versus targeted and total. Fine-tuning degrades capabilities its data never threatened [Luo et al., 2023, Kotha et al., 2024], LoRA forgets less than full fine-tuning [Biderman et al., 2024], and roughly 1 % rehearsal of prior data prevents much of it [Scialom et al., 2022] — the closest analogue to our dose result, and a reason to expect a small one to work. The difference is shape. Forgetting is measured as a spread-out decline across unrelated benchmarks; here the inverse direction goes to near zero while general probes move a few points, and the REPLAY arm exists to tell the two apart by substituting generic instruction data for the reversed pairs at identical budget.

Suppression rather than removal. Fine-tuning has been characterised as a thin wrapper over pretrained capability [Jain et al., 2024] and as suppressing rather than removing behaviour [Lee et al., 2024], and damaged capabilities can be recovered post hoc by filtering the weight update itself [spe, 2026]. lat [2026] add a caution we take seriously: objectives that repair reversal behaviour need not build a single direction-agnostic representation, and their probes are consistent with forward

and reverse stored as distinct entries. That is why §7 uses four instruments rather than one, and why both directions share a single system prompt — if directions can be stored separately while behaviour looks bidirectional, two personas would let disjoint circuits masquerade as a unified one.

Direction as a confound in attribution. The methodological consequence has a literature of its own: gains attributed to an objective often belong to the data mixture [Longpre et al., 2023], and reporting practice makes such claims hard to check [Lipton and Steinhardt, 2018]. Chen et al. [2025] is the sharpest current instance, and a useful one because nothing about it is careless. RevThink augments each example with a teacher-generated backward question and backward reasoning and improves *forward* accuracy by 13.53% across 12 datasets. Its objective is $\mathcal{L} = \frac{1}{3n} \sum_i [\ell(\text{forward}) + \ell(\text{backward question}) + \ell(\text{backward reasoning})]$, a sum of cross-entropy terms over a union of three formats — which is what supervised fine-tuning on that union is. Its baselines differ from it in data direction as well as in method. When an objective has that form, the objective and the data mixture are not separable by any experiment the paper contains. That is the general problem being structural rather than accidental, and it is why the test in §1 is worth stating rather than assuming.

10 Limitations

<<limitations-section>>

11 Conclusion

For any method that adds instances to paired training data, train the arm that supplies the same directional content and none of the method.

References

- The illusion of latent generalization: Bi-directionality and the reversal curse, 2026. Probes are consistent with forward and reverse stored as distinct entries rather than one direction-agnostic representation.
- Spectral unforgetting: Post-hoc recovery of damaged capabilities without retraining, 2026.
- Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. The reversal curse: Llms trained on “a is b” fail to learn “b is a”. In *International Conference on Learning Representations*, 2024.
- Dan Biderman, Jacob Portes, Jose Javier Gonzalez Ortiz, et al. Lora learns less and forgets less. *Transactions on Machine Learning Research*, 2024. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Justin Chih-Yao Chen, Zifeng Wang, Hamid Palangi, Rujun Han, Sayna Ebrahimi, Long T. Le, Vincent Perot, Swaroop Mishra, Mohit Bansal, Chen-Yu Lee, and Tomas Pfister. Reverse thinking makes llms stronger reasoners. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics*, 2025. Measures FORWARD accuracy: backward data added to improve forward reasoning. Loss is a sum of three cross-entropy terms over a union of formats (p. 4), and no baseline holds data direction fixed.

- Olga Golovneva, Zeyuan Allen-Zhu, Jason Weston, and Sainbayar Sukhbaatar. Reverse training to nurse the reversal curse, 2024.
- Eshgin Hasanov, Md. Mahadi Hassan Sibat, Shubhra Kanti Karmaker, and Aashish Yadavally. The path not taken: Duality in reasoning about program execution. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, 2026. DexBench: 445 paired instances, 13 LLMs evaluated. No fine-tuning. Code and data at <https://github.com/sail-ucf/dexbench>.
- Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, et al. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. In *International Conference on Learning Representations*, 2024. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Suhas Kotha, Jacob Mitchell Springer, and Aditi Raghunathan. Understanding catastrophic forgetting in language models via implicit inference. In *International Conference on Learning Representations*, 2024. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Andrew Lee, Xiaoyan Bai, Itamar Pres, et al. A mechanistic understanding of alignment algorithms: A case study on dpo and toxicity. In *International Conference on Machine Learning*, 2024. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Zachary C. Lipton and Jacob Steinhardt. Troubling trends in machine learning scholarship. *arXiv preprint arXiv:1807.03341*, 2018. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Shayne Longpre, Le Hou, Tu Vu, et al. The flan collection: Designing data and methods for effective instruction tuning. In *International Conference on Machine Learning*, 2023. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning, 2023. NOT YET READ – cited from the plan’s bibliography; verify before submission.
- Serge Lionel Nikiema, Jordan Samhi, Micheline Moumoula, Albarick Djié, Abdoul Kader Kaboré, Jacques Klein, and Tegawendé F. Bissyandé. Using contrastive learning to improve two-way reasoning in large language models: The obfuscation task as a case study. 2025. Names “cognitive specialization”: forward-only fine-tuning degrades pre-existing bidirectional reasoning. Declares a Bidirectional Fine-Tuning baseline (p. 6) and reports no number for it.
- Thomas Scialom, Tuhin Chakrabarty, and Smaranda Muresan. Fine-tuned language models are continual learners. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022. NOT YET READ – the 1 percent rehearsal result is the closest analogue to our dose finding and must be read before it is leaned on.
- Dawei Zhu, Pinzhen Chen, Miaoran Zhang, Barry Haddow, Xiaoyu Shen, and Dietrich Klakow. Fine-tuning large language models to translate: Will a touch of noisy data in misaligned languages suffice? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. URL <https://aclanthology.org/2024.emnlp-main.24/>. Single-direction fine-tuning generalizes, EXCEPT that English on the target side causes task misinterpretation damaging translation into non-English.